



macOS-Native Experience & CAD-Familiar Defaults

Making FreeCAD feel at home on Apple Silicon, and instantly familiar to SolidWorks and Fusion 360 users.

Repository **brucecodes2023/FreeCAD_BT**

Platform focus **macOS (arm64)**

Status **Scoping**

SUMMARY

This document scopes three workstreams, ordered by the priority you set: (A) make FreeCAD's *default* UI feel like a modern mechanical CAD tool in the SolidWorks / Fusion 360 tradition; (B) sharpen the *macOS-native* window experience; and (C) make FreeCAD *faster on Apple Silicon*. Each item below is grounded in the current codebase, notes what already exists versus what is new work, lists the files it touches, and rates effort by technical scope (not calendar time).

Two headline findings shape the plan:

- A **SolidWorks navigation style already exists** in-tree, so a familiar mouse/orbit model is mostly a *defaults* change, not new code.
- FreeCAD's main window **already uses the native macOS frame**, so the red/amber/green traffic lights are *already* rendered top-left by macOS. The real “native” work is unifying the toolbar with the title bar and de-styling non-native chrome — not moving the buttons.

Objectives & Priority

1. **CAD-familiar defaults first.** Out of the box, navigation, view cube, layout, and theme should read as a modern mechanical-CAD tool to a SolidWorks / Fusion user.
2. **Native macOS window chrome.** Title bar, toolbar, and panels should follow macOS conventions; verify the traffic-light placement is already correct.
3. **Apple Silicon performance.** Native arm64 build tuning, smarter startup, and a path toward a modern GPU backend.

Workstream A Priority 1 CAD-Familiar UI Defaults

Goal: a first launch that feels like SolidWorks / Fusion 360 without the user touching a single preference. The cleanest, lowest-risk vehicle is a built-in **Preference Pack** plus a small change to the shipped defaults.

A1 — Default to a familiar navigation model

Exists today. Navigation styles live under `src/Gui/Navigation/`; the current default is `CADNavigationStyle`. A `SolidWorksNavigationStyle` is *already implemented and registered* (via `src/Gui/SoFCDB.cpp`). The default is read from the parameter `BaseApp/Preferences/View→NavigationStyle`, with the code fallback set in `src/Gui/View3DSettings.cpp` and the template default in `src/Gui/PreferencePackTemplates/View.cfg`.

- **Change:** ship `NavigationStyle = Gui::SolidWorksNavigationStyle` as the default (via the new pack in A2, or the fallback in `View3DSettings.cpp`).
- **Effort:** Low. **Risk:** Low (reversible preference).

A2 — Ship a “Modern CAD” preference pack

New work, built on existing infrastructure. Preference packs live under `src/Gui/PreferencePacks/` and are managed by `Gui::PreferencePackManager` (`src/Gui/PreferencePacks/PreferencePackManager.cpp`). A pack is a folder with a `.cfg` (merged into user parameters), optional pre/post macros, and optional `.qss` stylesheet. Bundle into one pack:

- navigation style (A1) and view-cube configuration (A3);
- a toolbar/dock layout derived from `PreferencePackTemplates/Main_window_layout.cfg` (tree, properties, and a modeling-forward default);
- a light + dark stylesheet in the SolidWorks/Fusion visual register, alongside `src/Gui/Stylesheets/FreeCAD.qss`.

Effort: Medium. **Risk:** Low–Medium (layout/theme QA).

A3 — View cube tuned like a design tool

Exists today. The navigation cube (`src/Gui/NaviCube.cpp`, `src/Gui/Inventor/SoNaviCube.h`) is already a SolidWorks/Fusion-style view cube. It is fully configurable via `BaseApp/Preferences/–NaviCube` (corner, size, colors, fonts). Set sensible defaults (corner, size, subtle colors) inside the A2 pack. **Effort:** Low.

A4 — First-run onboarding hook

The Start workbench already has a first-run flow (`FirstStart2024` flag) with a theme selector: `src/Mod/Start/Gui/GeneralSettingsWidget.cpp` and `ThemeSelectorWidget`. Add the “Modern CAD” pack as a selectable (or default) option so new users land in the familiar experience. **Effort:** Low–Medium.

A5 — Stretch: a true Fusion 360 navigation style

New work. No `Fusion360NavigationStyle` exists. Fusion’s input model (MMB-orbit, Shift+MMB-pan, scroll-zoom) differs from SolidWorks. Add a `UserNavigationStyle` subclass under `src/Gui/Navigation/` and register it in `src/Gui/SoFCDB.cpp` + `src/Gui/CMakeLists.txt`. **Effort:** Medium. **Risk:** Low (additive).

A6 — Out of near-term scope: ribbon UI

Neither SolidWorks’ `CommandManager` nor Fusion’s toolbar is present; FreeCAD uses a classic `Gui::ToolBarManager` model (`src/Gui/ToolBarManager.cpp`) with a `WorkbenchSelector`. A ribbon is greenfield and large; recommend deferring in favor of A1–A4, which capture most of the “familiar” feel at a fraction of the risk.

Workstream B Priority 2 macOS-Native Window Chrome

B0 — Finding: traffic lights are already native & top-left

`Gui::MainWindow` (`src/Gui/MainWindow.cpp`) is a plain `QMainWindow` with default `Qt::Window` flags — no frameless or custom title bar, and no `setUnifiedTitleAndToolBarOnMac` anywhere in the tree. On macOS, Qt/Cocoa draws the standard frame, so the red/amber/green controls already sit top-left. If they appear top-right, the app is being viewed on Linux/Windows or through a themed build. **Action:** verify visually on macOS (Qt 6.8) and document; no code move is required to place the buttons.

B1 — Unified title + toolbar

Add `setUnifiedTitleAndToolBarOnMac(true)` in the `MainWindow` constructor to merge the top toolbar into the title bar, the hallmark of a native Mac app. Validate toolbar layout, full-screen, and the `ToolBarManager` areas. **Effort:** Low–Medium. **Risk:** Medium (layout regressions).

B2 — Native dock & panel chrome

Dock panels currently draw custom title bars via `QDockWidget::title` in `src/Gui/Stylesheets/FreeCAD.qss`, and floating toolbars use `Qt::FramelessWindowHint` (`src/Gui/ToolBarManager.cpp`). Platform-gate this QSS on macOS so Cocoa draws native dock/panel chrome. **Effort:** Medium. **Risk:** Medium (theme interplay).

B3 — Full-size content view (deep native integration)

For a truly modern Mac look (content under the title bar, large-title behavior), add `NSWindow` integration via a new Objective-C++ (.mm) shim. There is a precedent for macOS Obj-C in-tree (QuickLook under `src/MacAppBundle/QuickLook/`, 3Dconnexion `GuiNativeEventMac`), but *no* window-chrome Cocoa code exists yet. **Effort:** High. **Risk:** High (new native surface).

B4 — Bundle identity & appearance

The bundled `src/MacAppBundle/FreeCAD.app/Contents/Info.plist` ships an empty bundle identifier (filled by CMake `sed`); the conda recipe uses `org.freecad.FreeCAD` and sets `NSRequiresAquaSystemAppearance=False` (`package/rattler-build/osx/Info.plist.template`). Align a stable bundle id, dark-mode support, and icon (`freecad.icns`); track signing/notarization as a release concern. **Effort:** Low–Medium.

Workstream C Priority 3 Apple Silicon Performance

What already helps

Native arm64 comes from `conda-forge/pixi` (`osx-arm64` platform in `pixi.toml`); `ccache` is on by default (`CMakeLists.txt`); C++ workbench modules load lazily on activation; and hot paths (`mesh`, `TechDraw HLR`) already use `QtConcurrent`.

C1 — Pin architecture & deployment target

No `CMAKE_OSX_ARCHITECTURES` is set; the build follows the toolchain. Set `arm64` explicitly for reproducible native builds and confirm `CMAKE_OSX_DEPLOYMENT_TARGET` (package build defaults to 11.0). Files: `CMakePresets.json` (`conda-macos-*`), `package/rattler-build/build.sh`. **Effort:** Low.

C2 — Optimization: LTO/IPO & Apple CPU tuning

No LTO/IPO or `-mcpu=apple-m*` tuning in the main CMake. Add optional `INTERPROCEDURAL_OPTIMIZATION` and Apple-CPU flags for release builds (`cMake/FreeCAD_Helpers/SetGlobalCompilerAndLinkerSettings.cmake`). Measure binary size and link time. **Effort:** Medium. **Risk:** Medium (build-time/ABI QA).

C3 — Activate TBB parallelism

`tbb-devel` is already a dependency but `HAVE_TBB` is never defined, so the TBB code paths (e.g. `src/Mod/Import/App/ImportOCAF.cpp`) are dead. Wire `find_package(TBB) + HAVE_TBB` in `src/config.h.cmake` to light up parallel import/mesh paths on multi-core Apple Silicon. **Effort:** Medium. **Risk:** Medium.

C4 — Faster startup

All ~28 modules' `InitGui.py` scripts run eagerly at GUI startup (`src/Gui/FreeCADGuiInit.py`). Explore deferring non-essential workbench registration and leaning on the existing `BackgroundAutoloadModules` mechanism (`src/Gui/StartupProcess.cpp`) to cut time-to-interactive. **Effort:** Medium. **Risk:** Medium (ordering assumptions).

C5 — Long-term: modern GPU backend

Rendering is OpenGL-only through `Coin3D + QOpenGLWidget` (`src/Gui/Quarter/QuarterWidget.cpp`, `src/Gui/View3DInventorViewer.cpp`); macOS OpenGL is deprecated. A Metal path (via Qt RHI or MoltenVK) is a substantial `Coin + Quarter +`

viewer effort, not a toggle. Flag as strategic R&D, decoupled from A/B. **Effort:** Very High. **Risk:** High.

Sequencing

Phase	Focus	Items
Phase 1	Familiar defaults	A1, A3, A2 (pack), A4 onboarding
Phase 2	Native window feel	B0 verify, B1 unified toolbar, B4 bundle id
Phase 3	Native polish + speed	B2 dock chrome, C1, C3, C4
Phase 4	Strategic R&D	A5 Fusion nav, B3 NSWindow, C5 Metal

Validation Plan

Change	How it is proven
Default navigation / pack	Fresh profile launches with SolidWorks-style orbit and the Modern CAD layout; toggling back works.
View cube defaults	Cube appears in the configured corner with the intended size/-colors.
Traffic lights (B0)	Screenshot on macOS showing native top-left controls; documented.
Unified toolbar (B1)	Visual check of title/toolbar merge; full-screen and toolbar-area regression pass.
arm64 / LTO / TBB	Build reports target <code>arm64</code> ; benchmark import/mesh with and without TBB; record link time.
Startup (C4)	Time-to-interactive measured before/after, across representative workbenches.
No regressions	FreeCAD CLI + C++ self-tests remain green (as in the environment validation).

Grounded File Index

Concern	Key path(s)
Navigation styles (SolidWorks exists)	src/Gui/Navigation/*, src/Gui/SoFCDB.cpp
Navigation default & fallback	src/Gui/View3DSettings.cpp, PreferencePackTemplates/View.cfg
View cube	src/Gui/NaviCube.cpp, src/Gui/Inventor/SoNaviCube.h
Preference packs & manager	src/Gui/PreferencePacks/, PreferencePackManager.cpp
Stylesheets / theme	src/Gui/Stylesheets/FreeCAD.qss, defaults.qss
First-run onboarding	src/Mod/Start/Gui/GeneralSettingsWidget.cpp, ThemeSelectorWidget
Main window & chrome	src/Gui/MainWindow.cpp, src/Gui/ToolBarManager.cpp
macOS bundle	src/MacAppBundle/, package/rattler-build/osx/
Build presets / arch	CMakePresets.json, pixi.toml, package/rattler-build/build/build.sh
Compiler / linker flags	cMake/FreeCAD_Helpers/SetGlobalCompilerAndLinkerSettings.cmake
TBB gate	src/config.h.cmake, src/Mod/Import/App/ImportOCAF.cpp
Rendering stack	src/Gui/Quarter/QuarterWidget.cpp, src/Gui/View3DInventorViewer.cpp
GUI startup	src/Gui/FreeCADGuiInit.py, src/Gui/StartupProcess.cpp

Effort ratings describe technical scope (breadth of components and invasiveness), not calendar estimates. “Exists today” items are configuration/defaults changes; “new work” items add code behind existing extension points.